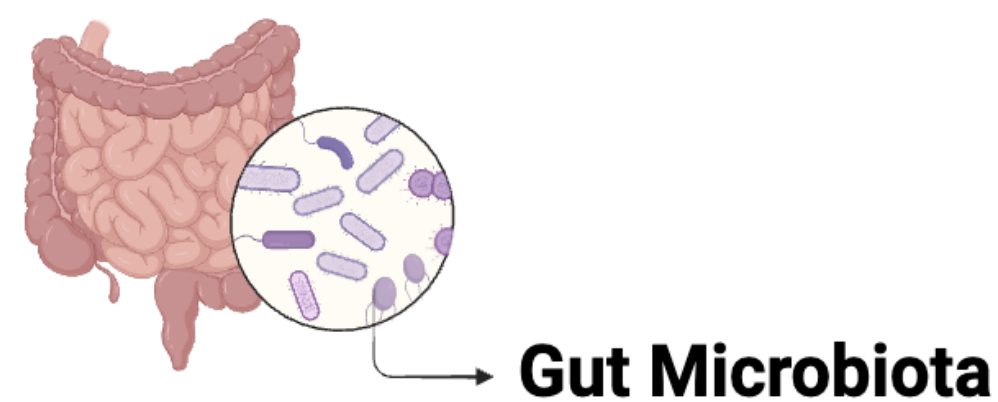


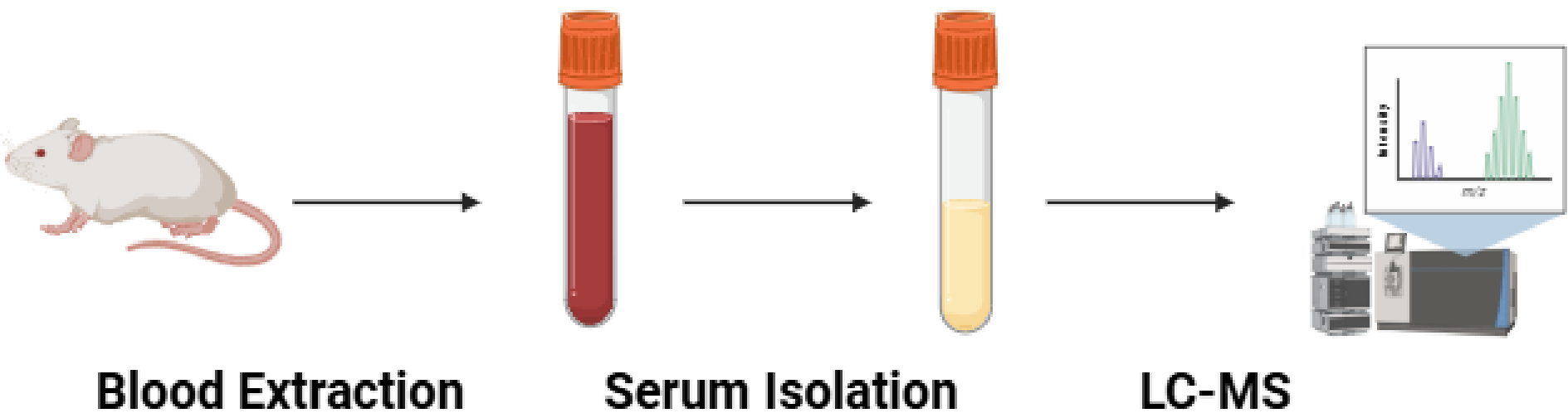
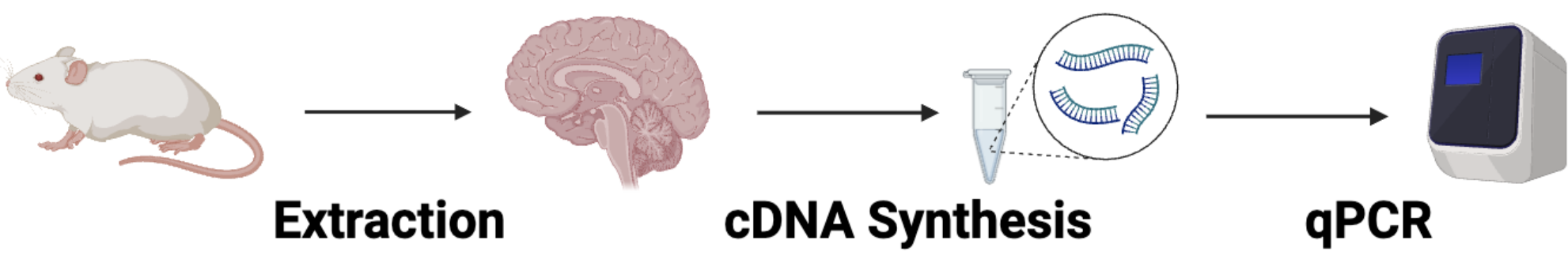
Background

- Inflammatory Bowel Disease (IBD) is a chronic condition characterized by an altered microbiome, causing inflammation of the gastrointestinal (GI) tract while impacting stress through HPA axis (hypothalamic-pituitary-adrenal) interactions
- People with IBD experience pubertal delay and neuropsychiatric impairments, both of which are associated with steroid hormone dysregulation and the “leaky” gut
- The specific mechanisms which cause people with IBD to experience these symptoms are currently unknown



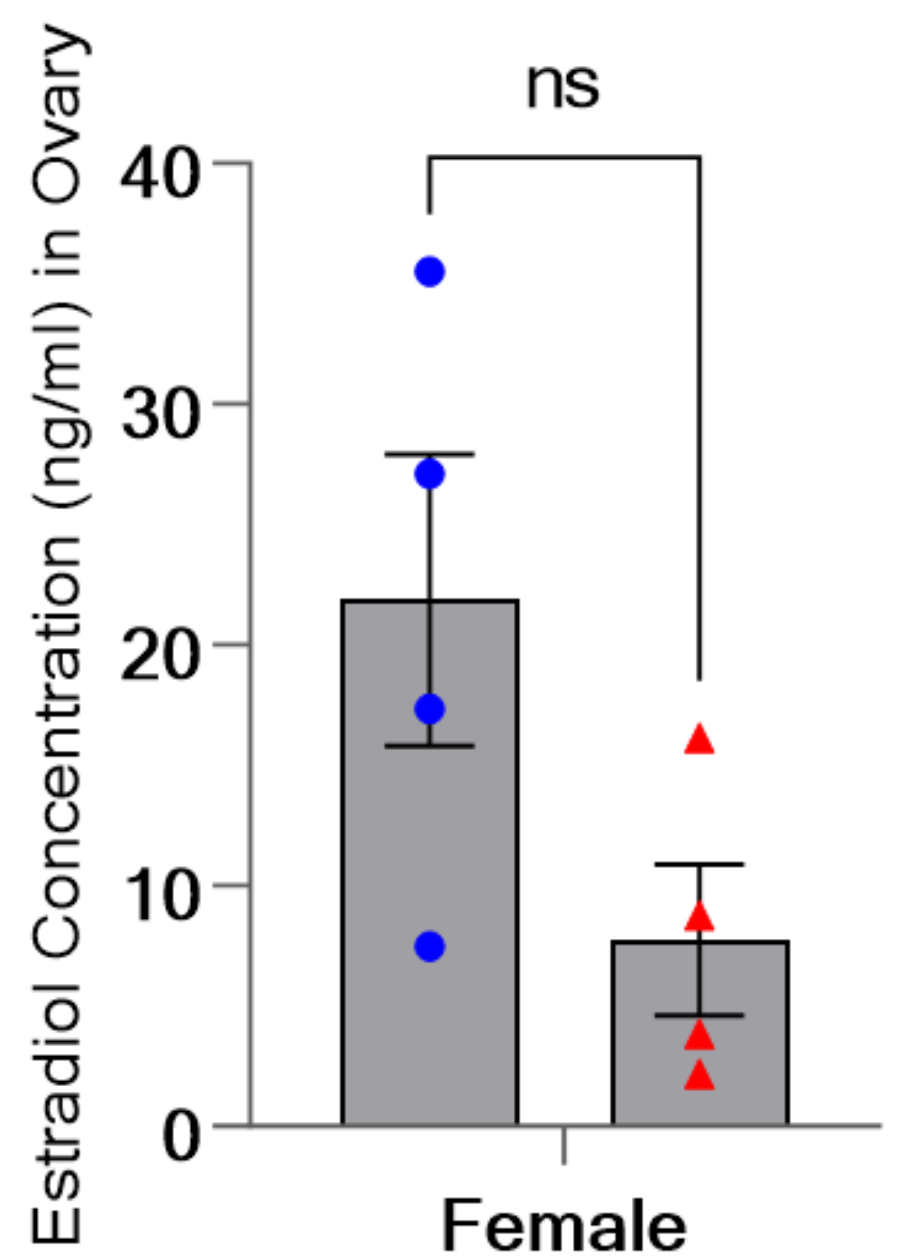
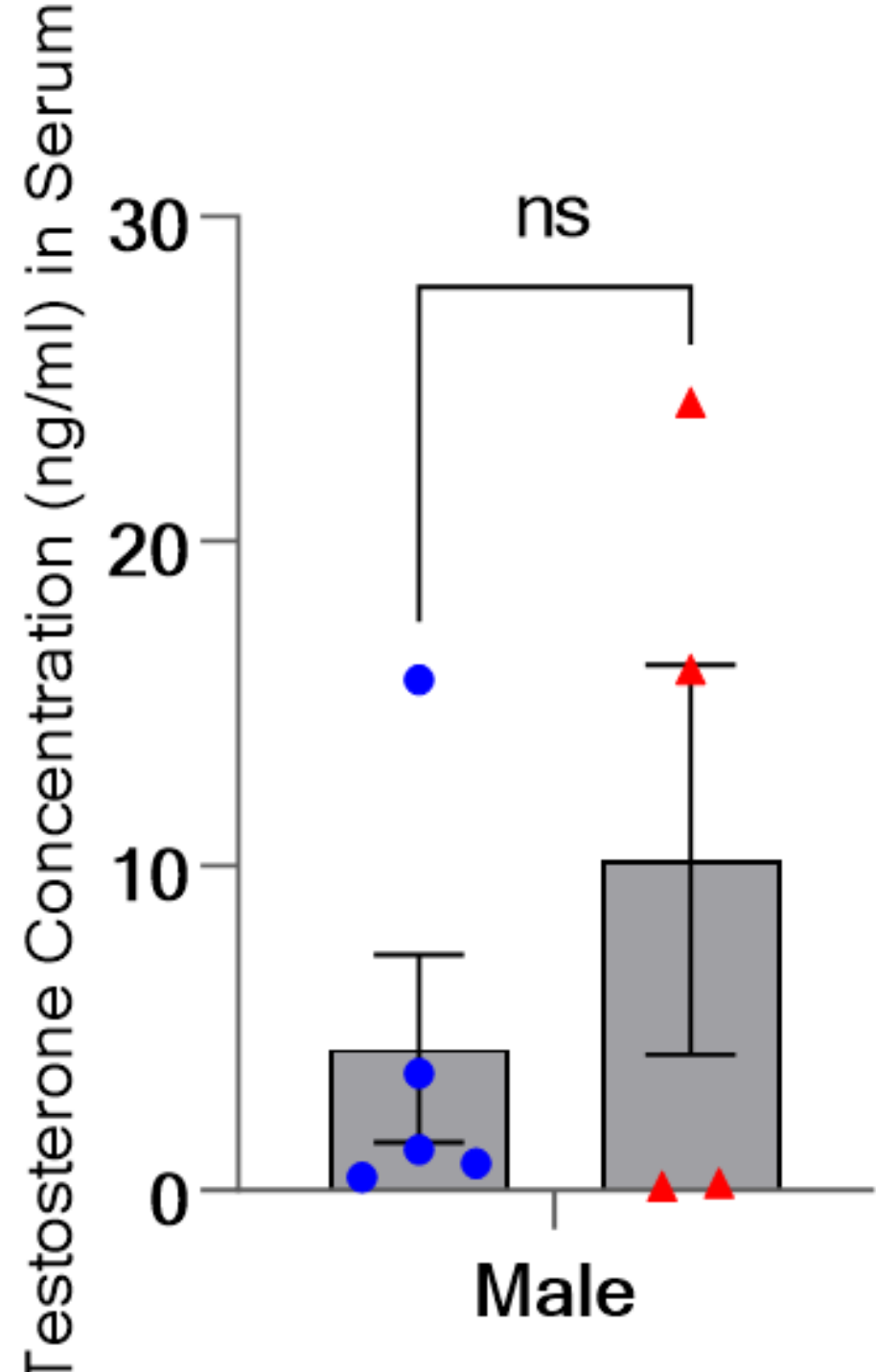
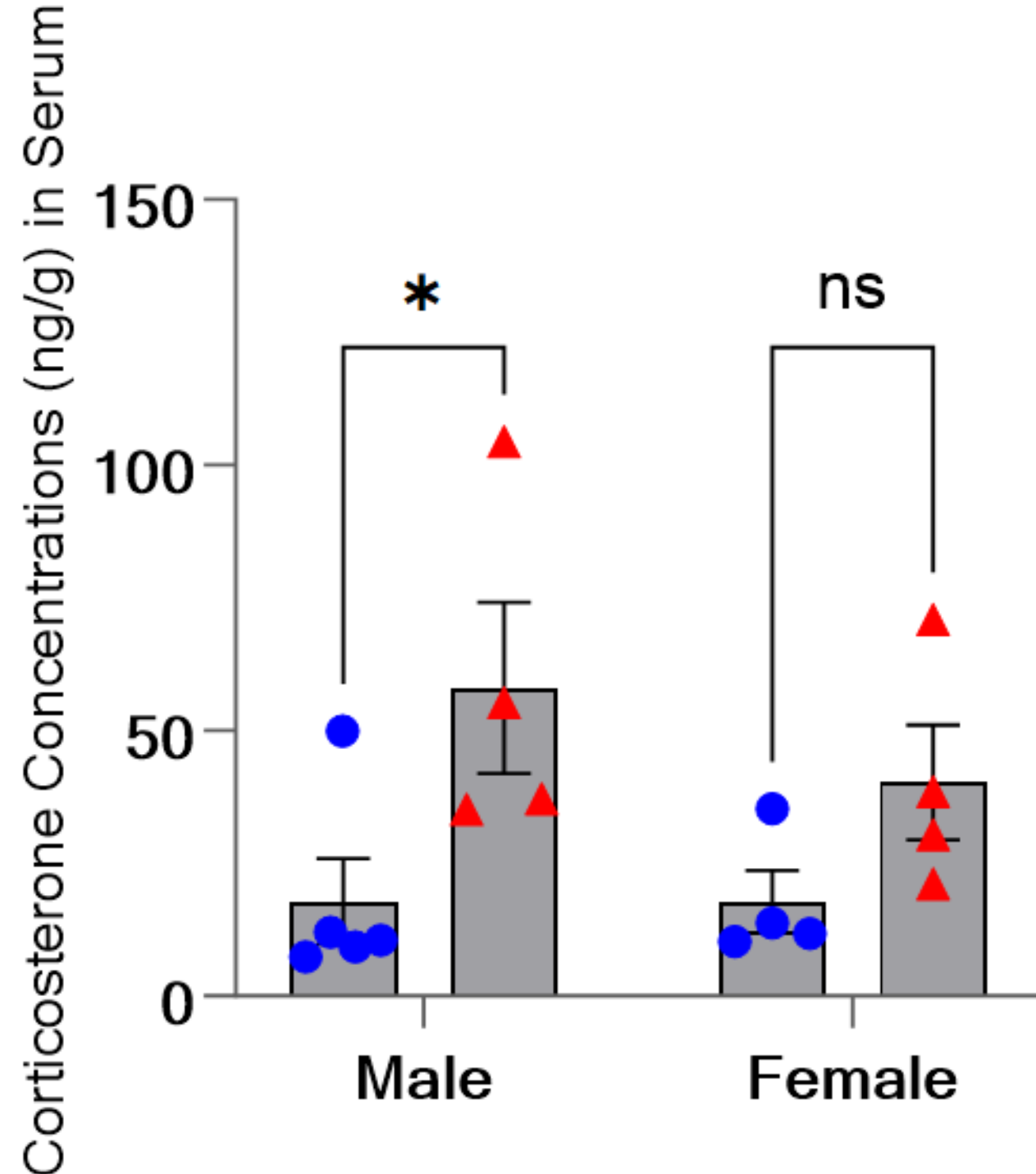
Methods

- Germ-free mice were gavaged with either a Healthy-Control (HC) or Inflammatory Bowel Disease (IBD) human microbiome
- Offspring of PND61 HC and IBD mice were euthanized, and their brains were collected for hypothalamic dissection and qPCR
- Blood was collected and centrifuged to collect serum. Ovary collection was needed to measure estradiol concentration
- Steroids were extracted via liquid-liquid extraction and analyzed by liquid-chromatography tandem mass-spectrometry

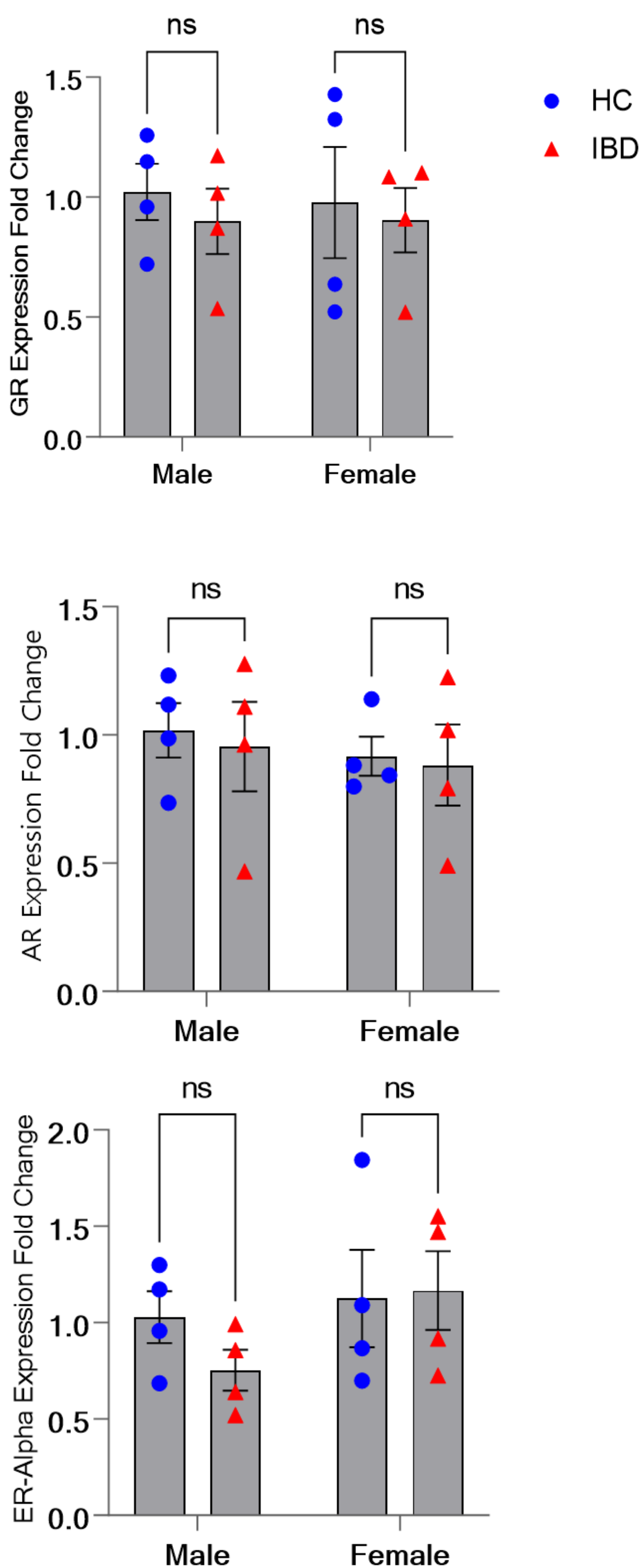


Results

HC vs IBD Steroid Levels



HC vs IBD Gene Expression



Acknowledgements

- Dr. Jordan Hamden, Dr. Annie Ciernia, and the Ciernia Lab provided excellent mentorship and support throughout this project at the DMCBH.
- The CIHR IRSC and Brain Canada generously funded the efforts behind this project.

Discussion

- In this first cohort, IBD mice surprisingly appear to possess higher levels of testosterone, and corticosterone in their serum than the HC mice.
- Presence of IBD gut microbiota increases the corticosterone and testosterone levels in male mice, while female mice show lower estradiol levels
- There was no change in hypothalamic gene expression, indicating that changes in steroid levels were not regulated by hypothalamic signaling
- IBD mice do not appear to have changes in gene expression relative to the HC mice, despite steroid levels appearing to differ – no correlation present



Future Directions

- Behavioural assays on reproductive activity to correlate gut-induced brain changes with functional outcomes regarding sex and hormone levels
- Behavioural assays on anxiety and depression, given that these behaviors are comorbid with IBD
- Metabolomic assays to analyze brain tissue or serum to detect metabolites derived from gut microbiota

